

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Built clean, modular, well-tested Python data pipelines with automated feature engineering and Advanced SQL-backed transforms, enforcing validation and observability, blocking 15% of failed releases.
- Engineered Kafka event-stream processing across distributed services for real-time data ingestion, handling missing, duplicate, and inconsistent data before it reaches downstream consumers.
- Built ML/data pipeline monitoring with error handling, logging, and drift-detection tooling replacing polling with event-driven observability, cutting redundant checks 90%.
- Set CI/CD pipelines for automated testing and deployment across the platform, optimizing pipeline performance on multi-tenant AWS and Azure infrastructure.

QureAI (AI Startup)

Mar 2026 – Present

ML Infrastructure Engineer

- Built PySpark data processing jobs handling medical imaging and other regulated healthcare data at scale, including cleansing, deduplication, standardization, and reconciliation for downstream training pipelines.
- Processed sensitive healthcare data (PHI) in a regulated product environment with security and compliance controls; built feature engineering and data-quality pipelines (pandas, NumPy) improving retrieval recall 61% to 89%.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production C++ and SQL-backed microservices in distributed systems for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Led data cleaning, preprocessing, and Advanced SQL storage redesign for nightly aggregation pipelines, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Standardized failure-handling contracts (retry, timeout, propagation semantics) across 25+ services and led incident response, reducing production escalations 60% in one quarter.
- Mentored engineers through code reviews and design write-ups, driving consistent engineering standards across the team.

Projects

LangFetch | tinyurl.com/LangFetch

- LangGraph agent compiling natural-language questions into dependency-ordered SQL subqueries with Pinecone vector retrieval and hybrid re-ranking; deployed as an MCP-compatible server for enterprise data queries.

Distributed KV Store / VPC Control Plane Simulator | github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus key-value store in C++ with leader election, log replication, and log compaction; companion VPC control plane simulator modeling subnet routing for fault-tolerant distributed coordination.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production LLM eval platform with hallucination detection, drift tracking, rubric-based scoring, and regression alerting across model versions; FastAPI backend, PostgreSQL.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, Advanced SQL, Java, TypeScript, C++

Data Engineering: PySpark, Kafka, ETL/ELT Pipelines, Data Cleansing/Validation/Deduplication, Feature Engineering

Multimodal & Compliance: Medical Imaging (Computer Vision), Healthcare/PHI Data Handling, Regulated Product Experience

Reliability & Cloud: CI/CD, Error Handling & Monitoring, AWS (SageMaker, Lambda, EC2), Azure, GCP, Docker

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer